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PTX10001-36MR L2 Circuit Scale

An HPE Juniper Networking Techpost

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1 PTX10001-36MR L2 Circuit Scale

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The PTX10001-36MR is well-known for its core, peering and DCI capabilities, but it's also a very performant L2 aggregation device. In this article, we will test and demonstrate L2 virtual circuits scale on Express4-based PTX routers.

1.1 Introduction

Each Express 4 ASIC (Codename: "BT") supports up to 16384 logical interfaces. In this test scenario we distribute logical interfaces between two PFEs of a PTX10001-36MR as shown in Figure 1:

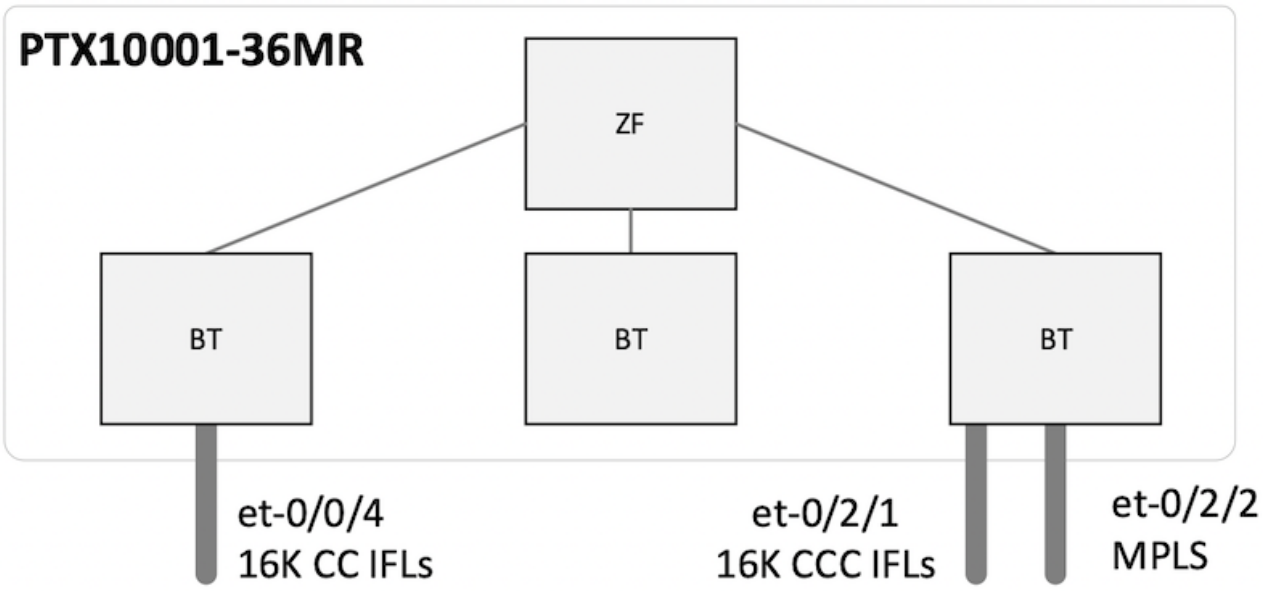


Figure 1: PTX10001 Internal Architecture

Figure 2 shows relationships between interfaces, circuits and remote edge devices. The setup has a trivial next-hop topology without any load balancing. Note that regular "et-" interfaces are used in this test. With Aggregated Ethernet in use the scale will reduce because, as of today, a separate logical interface is allocated for each member link (also known as "child" IFL).

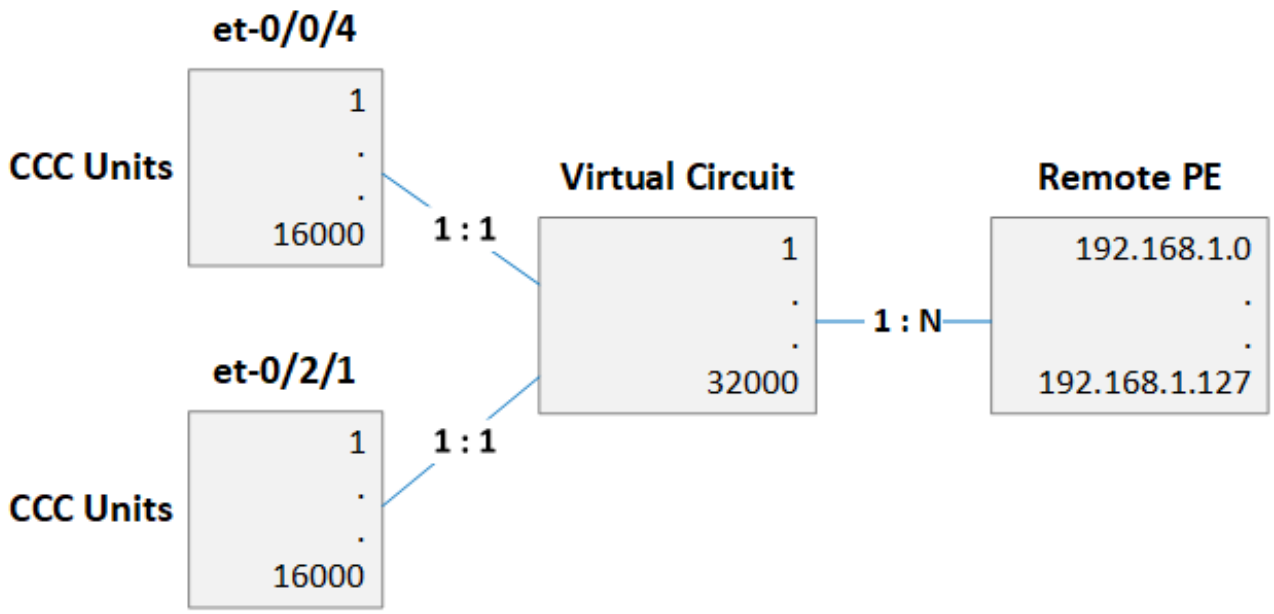


Figure 2: CCC Logical Interfaces, Circuits and Remote PEs.

1.2 Configuration and Validation

CCC interface configuration is shown below:

```
unit 1 {
   encapsulation vlan-ccc;
   vlan-tags outer 1001 inner 1;
   input-vlan-map {
      swap-swap;
      vlan-id 1;
      inner-vlan-id 100;
   }
   output-vlan-map swap-swap;
}
```

All the circuits are provisioned as static pseudo-wires (32000 total) and 128 static LSPs are provisioned to reach remote PEs.

All the l2 connections are “up” after commit:

```
regress@rtme-ptx10# run show l2circuit connections | match "rmt\s+Up" | count
Count: 32000 lines
```

PFE commands confirm that 32K MPLS and 32K CCC routes are present.

```
root@rtme-ptx10:pfe> show route summary

IPv4 Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%)
-------- ---------- ---------- --------- --------- ---------- ------
Default 35 3640 31 0 19 39
1 0 0 0 0 0 0
50 5 520 5 0 5 0
36738 5 520 5 0 5 0

MPLS Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%)
-------- ---------- ---------- --------- --------- ---------- ------
Default 32007 3328728 32007 0 32007 -

IPv6 Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%)
-------- ---------- ---------- --------- --------- ---------- ------
```

```
Default 7 728 7 0 7 0
1 0 0 0 0 0 0
50 5 520 5 0 5 0
36738 6 624 6 0 6 0

CLNP Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%)
-------- ---------- ---------- --------- --------- ---------- ------
Default 2 208 2 0 2 -
50 1 104 1 0 1 -

CCC Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%)
-------- ---------- ---------- --------- --------- ---------- ------
Default 32000 3328000 32000 0 32000 -
```

16K L2 circuits per PFE limit is imposed by the address space of the local L2 domain attribute of the ASIC (14 bits). The mapping of interfaces and VLANs to a logical interface is kept in the L2 domain table.

```
root@rtme-ptx10:pfe> show jexpr chash pfe 2
pfe : 2

  ID PFE Name Size(cells) Used(cells) % MaxUsed(cells) % Note
  -- --- ---- ----------- ----------- - -------------- - ----
   0 2 IGP-megidx 4096 0 0 0 0
   1 2 EGP-megidx 4096 0 0 0 0
   2 2 TunnelL2DomainHa 24576 0 0 0 0 1 entry=2/3/6 cells
   3 2 TunnelL2DomainHa 24576 0 0 0 0 1 entry=2/3/6 cells
   4 2 EPP_MAPID 196608 16120 8 16120 8
   5 2 L2Domain 32768 16012 48 16012 48
   6 2 SLU-MY-MAC 3072 39 1 39 1
   7 2 DLU-IDB 2519040 0 0 0 0 1 entry=1/2/4 cells
   8 2 SA-Learning 32768 0 0 0 0

  ID PFE Name Size(cells) Used(cells) % MaxUsed(cells) % Note
  -- --- ---- ----------- ----------- - -------------- - ----
   0 255 DLU-IDB 2519040 131287 5 415189 16 1 entry=1/2/4 cells
```

Utilization of ingress (next-hop) memory is less than 20%.

```
root@rtme-ptx10:pfe> show jexpr jtm ingress-main-memory chip 2
Show JTM Ingress MM allocation:
Pfe Chip: 2
[Global Region]
Used words: 129948, max used: 129948, allocs: 3792767, frees: 3754556
   Allocator 0: used words: 10825, max used: 10825, allocs: 316064, frees: 312877
      total words: 59392, available: 81%
   Allocator 1: used words: 10828, max used: 10828, allocs: 316064, frees: 312880
      total words: 63488, available: 82%
   Allocator 2: used words: 10826, max used: 10826, allocs: 316064, frees: 312880
      total words: 63488, available: 82%
   Allocator 3: used words: 10828, max used: 10828, allocs: 316064, frees: 312880
      total words: 63488, available: 82%
   Allocator 4: used words: 10832, max used: 10832, allocs: 316064, frees: 312880
      total words: 63488, available: 82%
   Allocator 5: used words: 10828, max used: 10828, allocs: 316064, frees: 312879
      total words: 63488, available: 82%
   Allocator 6: used words: 10826, max used: 10826, allocs: 316064, frees: 312880
      total words: 65536, available: 83%
   Allocator 7: used words: 10832, max used: 10832, allocs: 316064, frees: 312880
      total words: 65536, available: 83%
   Allocator 8: used words: 10832, max used: 10832, allocs: 316064, frees: 312880
      total words: 65536, available: 83%
   Allocator 9: used words: 10832, max used: 10832, allocs: 316064, frees: 312880
      total words: 65536, available: 83%
   Allocator 10: used words: 10834, max used: 10834, allocs: 316064, frees: 312880
      total words: 65536, available: 83%
   Allocator 11: used words: 10825, max used: 10825, allocs: 316063, frees: 312880
      total words: 65536, available: 83%
[Private Region]
Used words: 454, max used: 454, allocs: 521, frees: 5
   Allocator 0: used words: 454, max used: 454, allocs: 521, frees: 5
      total words: 4096, available: 88%
[Load Balance]
   Allocator 0: used words: 144, max used: 144, allocs: 7, frees: 1
     overflow, used words: 0, max used: 0, allocs: 0, frees: 0
      total words: 6144, available: 97%
[SLT]
```

```
    Allocator 0: used words: 0, max used: 0, allocs: 0, frees: 0
       total words: 32768, available: 100%
[RPL]
    Allocator 0: used words: 0, max used: 0, allocs: 0, frees: 0
       total words: 2048, available: 100%
[REF]
    Allocator 0: used words: 8, max used: 8, allocs: 1, frees: 0
       total words: 8192, available: 99%
```

Egress memory is used for encapsulation structures, such as output VLAN IDs, forwarding MAC addresses and MPLS labels. 4 partitions out of 12 are in use and utilization of these 4 is less than 70%. Overall utilization is less than 30%.

```
root@rtme-ptx10:pfe> show jexpr jtm egress-memory summary chip 2
Show JTM Egress Memory Summary:
Pfe Chip: 2
 Max Total Encap Memory Used (across all partitions in words) : 86155
 Current Total Encap Memory Used (across all partitions in words) : 86155

 PartitionID CurrType Total(words) Used(words) % MaxUsed(words) %

 ----------- -------- ------------ ----------- -- -------------- --

    0 MacTbl 32768 0 0 0 0

    1 UnusedPool 32768 0 0 0 0

    2 UnusedPool 32768 0 0 0 0

    3 UnusedPool 32768 0 0 0 0

    4 Public 32768 18680 57 18680 57

    5 Public 32768 30720 93 30720 93

    6 UnusedPool 32768 0 0 0 0

    7 Public 32768 30720 93 30720 93

    8 Private 32768 6035 18 6035 18

    9 UnusedPool 32768 0 0 0 0

   10 Private 32768 0 0 0 0

   11 Tunnel 4096 0 0 0 0
```

1.3 Configuration Generation Script

```
import sys
import ipaddress

left_ifd = 'et-0/0/4'
right_ifd = 'et-0/2/1'
mpls_ifd = 'et-0/2/2'
first_nbr = ipaddress.IPv4Address( "192.168.1.0")
ifl_count = 16000
nbr_count = 128

def generate_config():
   print("interfaces {")
   print("\t{:s} {{".format(left_ifd))
   print("\t\tflexible-vlan-tagging;")
   print("\t\tencapsulation flexible-ethernet-services;")
   for i in range(0, ifl_count):
      ivlan_left = i % 4000 + 1
      ovlan_left = i // 4000 + 1
      print( "\t\tunit {:d} {{".format( i+1))
      print( "\t\t\tencapsulation vlan-ccc;")
      print( "\t\t\tvlan-tags outer {:d} inner {:d};".format(ovlan_left, ivlan_left))
      print( "\t\t\tinput-vlan-map {")
      print( "\t\t\t\tswap-swap;")
      print( "\t\t\t\tvlan-id 1;")
      print( "\t\t\t\tinner-vlan-id 100;")
```

```
        print( "\t\t\t}")
        print( "\t\t\toutput-vlan-map swap-swap;")
        print( "\t\t\t}")
    print("\t}")
    print("\t{:s} {{".format(right_ifd))
    print("\t\tflexible-vlan-tagging;")
    print("\t\tencapsulation flexible-ethernet-services;")
    for i in range(0, ifl_count):
        ivlan_right = i % 4000 + 1
        ovlan_right = i // 4000 + 1 + 1000
        print( "\t\tunit {:d} {{".format( i+1))
        print( "\t\t\tencapsulation vlan-ccc;")
        print( "\t\t\tvlan-tags outer {:d} inner {:d};".format(ovlan_right, ivlan_right))
        print( "\t\t\tinput-vlan-map {")
        print( "\t\t\t\tswap-swap;")
        print( "\t\t\t\tvlan-id 1;")
        print( "\t\t\t\tinner-vlan-id 100;")
        print( "\t\t\t}")
        print( "\t\t\toutput-vlan-map swap-swap;")
        print( "\t\t}")
    print("\t}")
    print("}")
    print("protocols {")
    print("\tl2circuit {")
    last_nbr_id = -1
    vc_id = 1
    for ifd in (left_ifd,right_ifd):
        for i in range(0, ifl_count):
            nbr_id = i // nbr_count
            if nbr_id!=last_nbr_id:
                if( last_nbr_id >= 0):
                    print("\t\t}") #close nbr container
                print("\t\tneighbor {:s} {{".format( str(first_nbr+nbr_id)))
            print("\t\t\tinterface {:s}.{:d} {{".format(ifd,i+1))
            print("\t\t\t\tstatic {")
            print("\t\t\t\t\tincoming-label {:d};".format(1000000+vc_id))
            print("\t\t\t\t\toutgoing-label {:d};".format(1000000+vc_id))
            print("\t\t\t\t}")
            print("\t\t\t\tvirtual-circuit-id {:d};".format(vc_id))
            print("\t\t\t}")
            last_nbr_id = nbr_id
            vc_id += 1
    if( last_nbr_id >= 0):
        print("\t\t}") #close nbr container
    print("\t}")
    print("\tmpls {")
    for nbr_id in range(0,nbr_count):
        print("\t\tstatic-label-switched-path p{:d} {{".format( nbr_id))
        print("\t\t\tingress {")
        print("\t\t\t\tnext-hop 192.168.0.2;")
        print("\t\t\t\tto {:s};".format( str(first_nbr+nbr_id)))
        print("\t\t\t\tpush {:d};".format( 1000000+nbr_id))
        print("\t\t\t}")
        print("\t\t}")
    print("\t}")
    print("}")

# execute
if __name__ == "__main__":
    generate_config()
else:
    pass
```

1.4 Glossary

- CCC: Circuit Cross-Connect
- IFL: Logical Interface
- LSP: Label Switched Path
- PE: Provider Edge
- PFE: Packet Forwarding Engine



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